class11

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Cooperative analysis with PCA

First step find and ADK sequence of :

```
library(bio3d)
id <- "1ake_A"  ## Change this to fun a different analysis
aa <- get.seq( id)
```

```
Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.
```

```
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214
```

```
Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

next step, is search the PDB database foor all related entries:

```
#blast <- blast.pdb(aa)
#hits <- plot (blast)
```

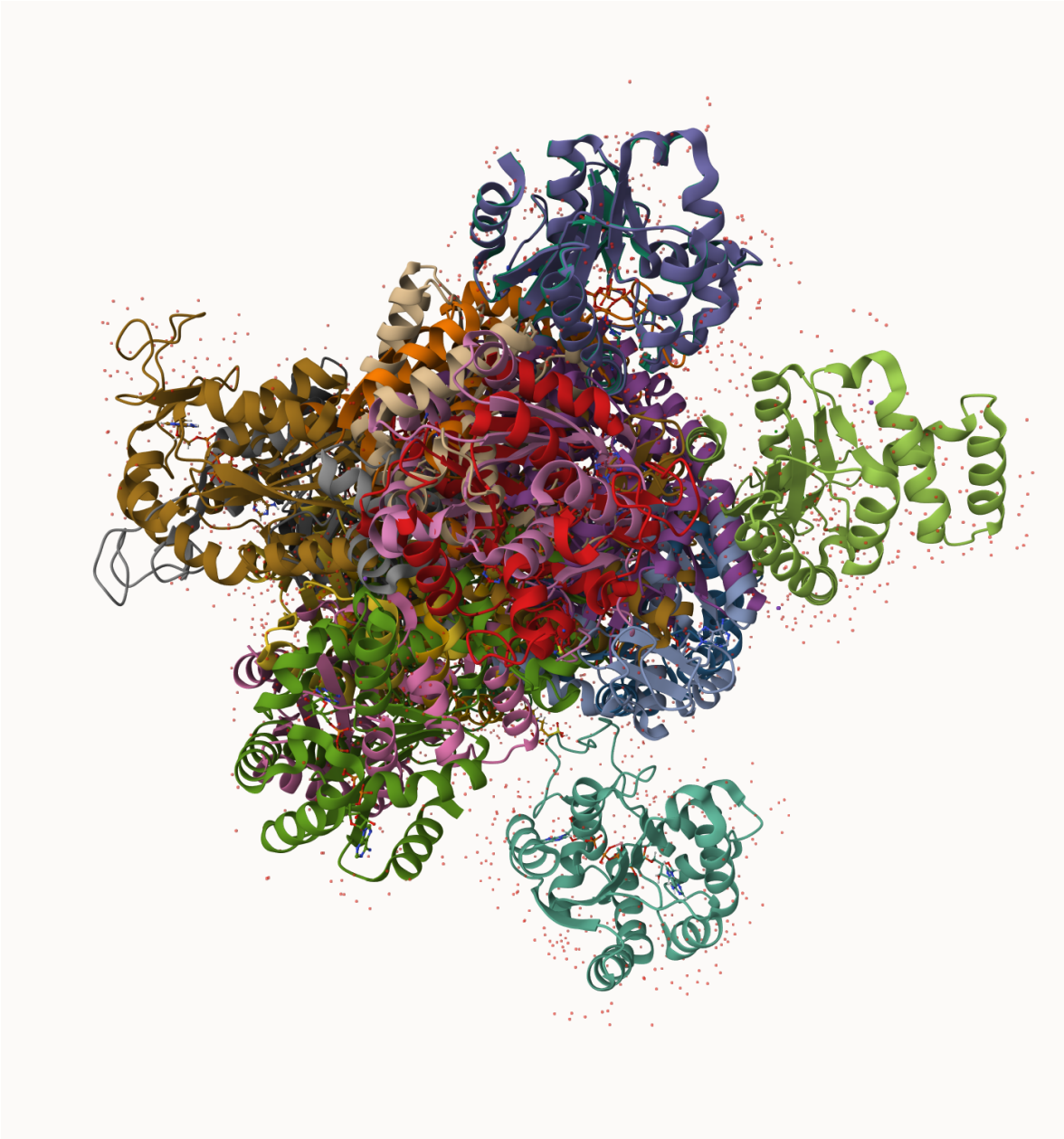
ALL the Blast results are here

```
#head (blast $hit.tbl )
```

The "top hits" are in the `hits` object. Now we can download these to our computer. Put these in a wee sub-folder (directory) called "pdbs" and use gzip tp speed things up

```
# Download releated PDB files
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Error:
! object 'hits' not found
```



We get a hot mess:

Next we will use the `pdalbn()` function to laign optionally fit (i.e. superpose) the identified PDB structures

This requires a BioConductor package called "msa" that we need to install. First we install BiocManager. Then we use `BiocManager::install("msa")`in

```r
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Error:
! object 'files' not found
```

We could view these in R with **bio3dview** `view.pdbs()` function

```r
library(bio3dview)
view.pdbs(pdbs, colorScheme="residue")
```

```
Error:
! object 'pdbs' not found
```

PCA

We can run PCA on our `pdbs` object using the `pca()` function form **bio3d**:

```r
pc.xray <- pca(pdbs)
```

```
Error:
! object 'pdbs' not found
```

```r
plot(pc.xray)
```

```
Error:
! object 'pc.xray' not found
```

```r
plot(pc.xray, 1:2)
```

```
Error:
! object 'pc.xray' not found
```

We can make a visualilzation of the major conformational difference (i.e. large sclae structure change) captured by our PCA analysis with the `mktrj()` function

```r
pc1 <- mktrj(pc.xray, file="pca.pdb")
```

```
Error:
! object 'pc.xray' not found
```

Let's see in Mol-star

Background

we saw last day that the main repository for biomolecular structure (the PDB datatbase) only has ~250,000 entries

UnitProtKB (the main protein sequence database) has over 200 million entries!

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al, 2021),

Without the aid of such approaches, it can take years of expensive Laboratory work to determine the structure of just one protein. With Alphafold me can now accurately compute a typical protein structure in as little as ten minutes.

The EBI AlphaFold database

The EBI database contains lots of computed structure models. it is increasingly likelky that the structure you are interest in is already in this database at https://alphafold.ebi.ac.uk >

There are 3 major outputs form ALphaFold

1. A model of structure in **PDB** format.
2. a **PLDDT score**: that tells us how confident the model is for a given residue in your protein (High values are good, above 70)
3. a **PAE score** that tells us about protein packing quality

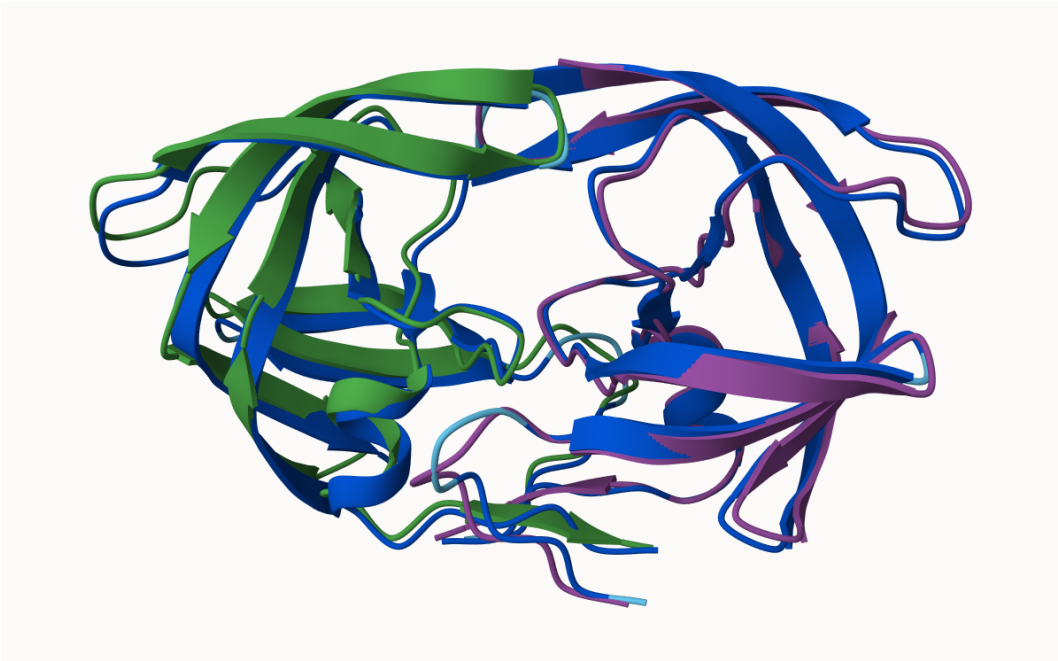
If you can't find a matching entry for the sequnec you ar einterested in AFDB you can run AlphaFold yourself…

Running AlphaFold

We will use ColabFold to run AlphaFold on our sequence < https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb

Figure form AlphaFold here!

```
library(png)
library(grid)

img <- readPNG("HIVPR_23119.png")
grid.raster(img)
```



Interpreting Results

Custom ananlysis of resulting models

We can read all the Alpha Fold rsults into R and do more quantitative analysis than just viewing the structure in Mol-star:

Read all the PDB models:

```r
library(bio3d)
p <- read.pdb("hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed
```

```r
results_dir <-list.files("hivpr_23119/", pattern= ".pdb", full.names =T)
```

```r
results_dir <- "hivpr_23119/"
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
print(pdb_files)
```

```
[1] "hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
[2] "hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
[3] "hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
[4] "hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
[5] "hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

```r
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

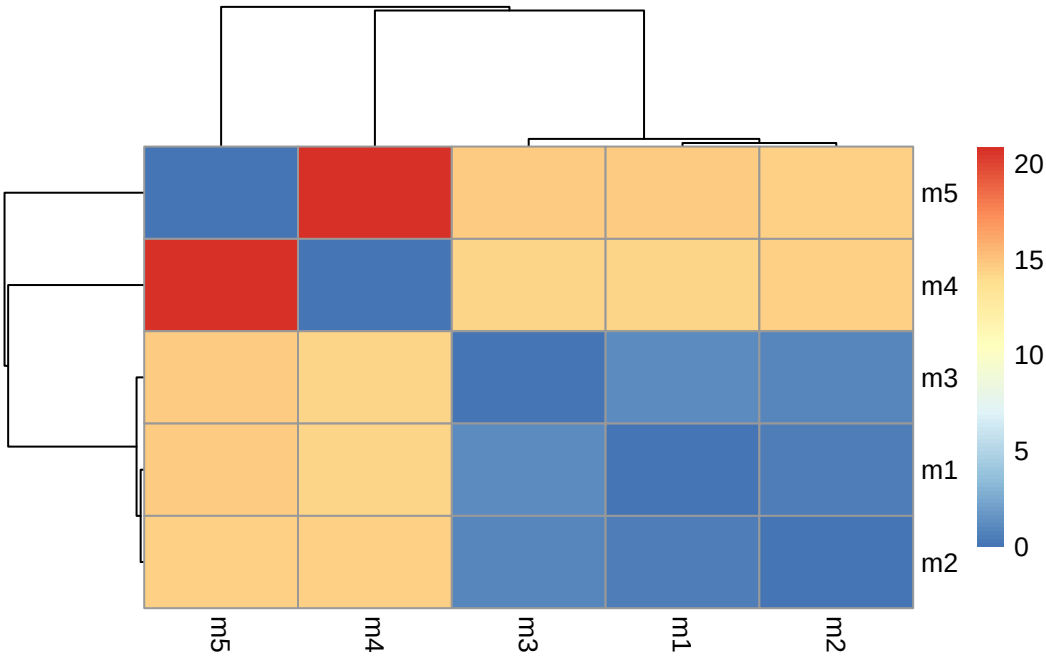
```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_
```

How similar or different are my models?

```r
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 198 non NA positions
```

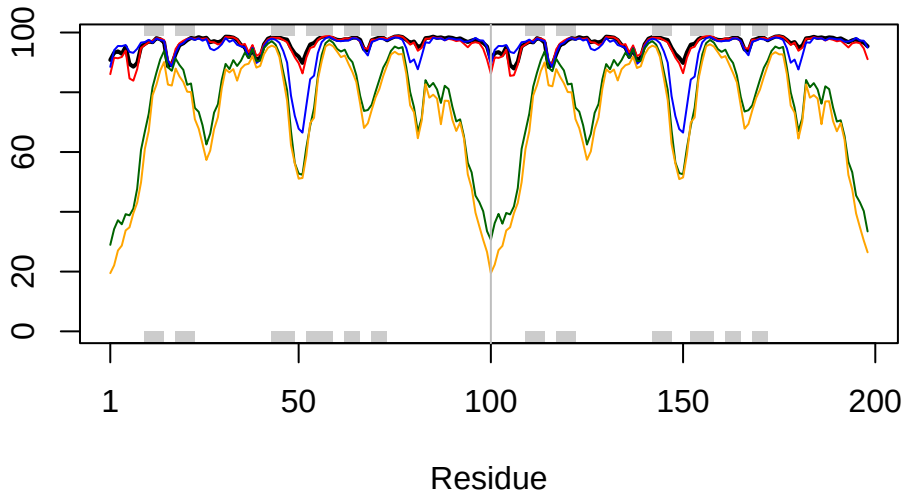
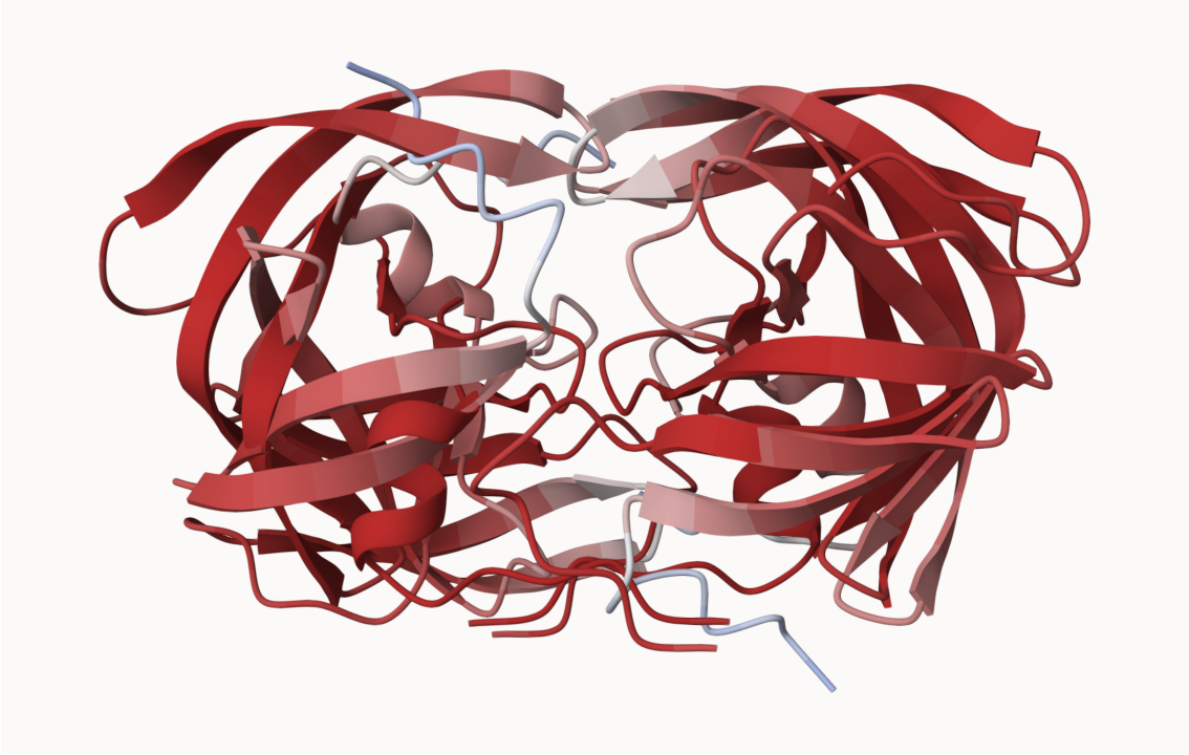
```r
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```


100
60
20
0
1
50
100
150
200
Residue